Pearls AQI Predictor

A 3-Day Air Quality Index Forecasting System

Internship Project Report

INTERN DETAILS

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1. Introduction

Air pollution is a critical environmental challenge affecting urban health and quality of life. The Air Quality Index (AQI) quantifies air pollution levels and forecasts their impact.

The **"Pearls AQI Predictor"** project develops an automated system to predict AQI for the next three days using historical and real-time data. It integrates data pipelines, machine learning models, and a serverless architecture to generate accurate forecasts. The system also includes model explainability using SHAP and an interactive dashboard with alerts for hazardous air quality, demonstrating practical data science and MLOps applications for real-world problems.

2. Problem Statement

Urban air pollution poses serious health risks and affects the overall quality of life. Predicting air quality manually is challenging due to rapidly changing environmental conditions and multiple influencing factors like weather and pollutant levels. Existing forecasting systems often lack automation, real-time updates, and interpretability.

The goal of this project is to build an **automated, scalable, and explainable system** that predicts the Air Quality Index (AQI) for the next three days. The system integrates historical and real-time data, applies machine learning models, and provides actionable insights through an interactive dashboard with alerts for hazardous conditions.

3. Project Objectives

- Collect and process historical and real-time air quality and weather data from external APIs.
- Engineer meaningful features such as time-based variables and AQI change rates for accurate predictions.
- Train and evaluate multiple machine learning and deep learning models using metrics like RMSE, MAE, and R^2(for Regression) & Accuracy and F1 for classification.
- Automate feature pipelines and model training workflows for continuous updates.
- Develop an interactive web dashboard to display real-time and forecasted AQI with alerts for hazardous levels.
- Implement model explainability techniques (e.g., SHAP) to provide insights into key factors affecting air quality.

4. System Architecture Overview

The Pearls AQI Predictor is designed as an end-to-end, automated machine learning pipeline that integrates real-time environmental data, performs feature engineering, trains predictive models, and delivers explainable air quality forecasts through an interactive dashboard.

1. Data Acquisition and Integration

Overview

This project collects historical and real-time air quality and weather data from external APIs. AQI data is obtained from OpenWeather, while meteorological data is sourced from Open-Meteo. Both datasets are aligned on an hourly timestamp and integrated into a unified dataset for downstream feature engineering and model training.

I. AQI Data (OpenWeather)

The OpenWeather Air Pollution API provides hourly air quality components. Duplicate timestamps are removed, and values are cleaned before integration.

Columns:

timestamp, pm2_5, pm10, no2, o3, co, so2

II. Weather Data (Open-Meteo)

Hourly weather data is collected from the Open-Meteo archive API, capturing key atmospheric conditions that influence air pollution behavior.

Columns:

timestamp, temperature, humidity, pressure, wind_speed, wind_direction, precipitation

Time Alignment and Integration

Both datasets are reindexed to a full hourly timeline, missing values are handled using interpolation and forward filling, and the data is merged on the timestamp to create a consistent, analysis-ready dataset.

2. Feature Engineering

The feature engineering pipeline transforms cleaned AQI and weather data into meaningful predictors that capture temporal patterns, pollutant behavior, and AQI dynamics, making the dataset suitable for robust forecasting.

1. AQI Calculation Features

- **AQI Numeric (aqi)**

Calculated using US EPA breakpoint standards by computing sub-indices for each pollutant and taking the maximum value.

Justification: Provides a standardized and interpretable AQI value used as the primary prediction target.

- **AQI Class**

AQI values are mapped into categorical classes (Good → Hazardous).

Justification: Enables multi-horizon AQI classification (risk-level prediction).

2. Time-Based Features

hour, day_of_week, month, is_weekend

Cyclical encodings: hour_sin, hour_cos, dow_sin, dow_cos

Justification:

Air pollution follows strong daily, weekly, and seasonal cycles. Cyclical encoding preserves the periodic nature of time variables and improves model learning.

3. Lag Features

AQI lags: aqi_lag_1h, 3h, 6h, 12h, 24h, 48h, 72h

Pollutant lags: lagged values of pm2_5, pm10, no2, o3, co, so2

Weather lags: lagged temperature, humidity, pressure, wind variables

Justification:

AQI is highly dependent on past pollution levels and weather conditions. Lag features allow the model to learn temporal dependencies and delayed effects.

4. Rolling Window Features

aqi_roll_mean_6h, 12h, 24h

aqi_roll_std_12h, 24h

Justification:

Rolling statistics capture short-term and daily trends, smooth noise, and represent pollution stability or volatility over time.

5. Delta (Change) Features

aqi_delta_1h, 3h, 6h, 24h

Justification:

These features measure the rate of change in AQI, helping the model detect rapid pollution increases or decreases.

6. Ratio Features

pm2_5_pm10_ratio

no2_o3_ratio

Justification:

Ratios capture chemical and source-related relationships between pollutants, providing insight beyond absolute concentrations.

7. Future Target Generation (Training Mode)

a) target_aqi_t_plus_24h
b) target_aqi_class_t_plus_24h
c) target_aqi_class_t_plus_48h
d) target_aqi_class_t_plus_72h

Justification:

Enables multi-horizon forecasting, allowing the system to predict AQI values and risk categories up to 3 days ahead.

Summary

This feature set combines temporal context, historical dependence, statistical trends, and pollutant interactions, ensuring the model receives rich, informative inputs for accurate AQI forecasting.

3. Model Training

The model training pipeline uses engineered features to forecast AQI at multiple future horizons. Air quality data is collected from OpenWeather, where AQI is primarily available as AQI classes (1 to 5). To handle this:

- **Classification** is used for Day-2 (+48h) and Day-3 (+72h) predictions.

- Hybrid approach is applied for Day-1 (+24h), where a regression model is additionally trained to predict numeric AQI values, even though the source data is categorical. Day-1 regression worked correctly, but Day-2 and Day-3 regression results were unreliable, so classification is used for longer horizons.

Trained Regression Models (Day-1)

- HistGradientBoosting
- GradientBoosting
- RandomForest

Trained Classification Models (Day-1, Day-2, Day-3)

- RandomForest
- GradientBoosting
- XGBoost

Multiple models are trained using a time-aware train–test split, and the best model for each target is selected based on standard evaluation metrics. All trained models are versioned and stored in MongoDB Atlas, with the active model used directly for prediction.

DAY 1

Best Regression Model (Day-1 Numeric AQI, +24h)

Model Name	Version	RMSE	MAE	R^2 Score
HistGradientBoosting	v1	37.89	22.54	0.235

HistGradientBoosting gave the best numeric AQI predictions for Day-1, with the lowest RMSE and highest R^2 among tested models. The R^2 indicates moderate predictive power, suitable for short-term regression forecasting.

Best Classification Models (AQI Class Prediction)

Day-1 Classification (+24h)

Model Name	Version	Accuracy	F1 (Weighted)
Random Forest	v1	0.7864	0.7564

Random Forest performed best for Day-1 AQI class prediction. High weighted F1 score shows it balances precision and recall well across all AQI classes, making it reliable for short-term risk categorization.

DAY 2

Day-2 Classification (+48h)

Model Name	Version	Accuracy	F1 (Weighted)
Gradient Boosting	v1	0.6755	0.6110

Gradient Boosting is optimal for Day-2 classification. As forecasting horizon increases, model performance drops slightly, reflecting growing uncertainty in medium-term predictions.

DAY 3

Day-3 Classification (+72h)

Model Name	Version	Accuracy	F1 (Weighted)
XGBoost	v1	0.6594	0.5690

XGBoost achieves the best Day-3 classification. Lower scores show long-term AQI prediction is inherently more challenging, but it still provides actionable risk-level forecasts.

4. CI/CD Automation Using GitHub Actions

Two automated workflows are implemented using **GitHub Actions** to ensure continuous data updates, model training, and prediction.

1. Feature Pipeline

Purpose:

Continuously updates raw data, cleans it, and generates engineered features**.**

Workflow Tasks:

- Ingests the latest AQI and weather data
- Builds a clean dataset
- Performs feature engineering
- Applies feature sanity checks

Execution Schedule:

- Runs every hour (24 times per day)
- Can also be triggered manually

Outcome:

Keeps the Feature Store up to date with the most recent data for real-time predictions.

2. Train + Predict Pipeline

Purpose:

Trains AQI forecasting models and generates multi-day predictions.

Workflow Tasks:

- Trains regression and classification models
- Selects and registers the best-performing models
- Generates AQI predictions using active models

Execution Schedule:

- Runs once per day (daily) at a scheduled time
- Can also be triggered manually

Outcome:

Ensures models are regularly retrained on updated data and fresh predictions are generated.

Summary

The hourly Feature Pipeline ensures continuous data freshness, while the daily Train + Predict Pipeline maintains model accuracy and produces reliable AQI forecasts in an automated, production-ready manner.

5. Web Dashboard (Streamlit Application)

An interactive web dashboard is developed using Streamlit to visualize AQI conditions, forecasts, trends, and model insights. The dashboard consists of four dedicated pages, each serving a specific analytical purpose.

Page 1: AQI Forecast Dashboard (Main Page)

This page presents:

- Current day AQI status
- Next 3-day AQI predictions
- Day 1 (+24h): Numeric AQI (regression) + AQI class
- Day 2 (+48h) & Day 3 (+72h): AQI class predictions only
- Supporting weather indicators (temperature, pressure, humidity, wind)

This design reflects the modeling strategy where short-term AQI is predicted numerically, while longer horizons rely on classification-based forecasts.

Page 2: AQI Trends & History

This page visualizes:

- Last 48 hours AQI trend
- 3-hour moving average
- AQI category distribution

It helps analyze recent air quality behavior and short-term variability using trend and distribution plots.

Page 3: Model Registry Dashboard

This page provides:

- A model governance view showing all trained models
- Model versioning, training date, performance metrics
- Clear distinction between active and archived models

It supports transparency, traceability, and lifecycle management of ML models.

Page 4: SHAP Explainability (Day-1 Regression Only)

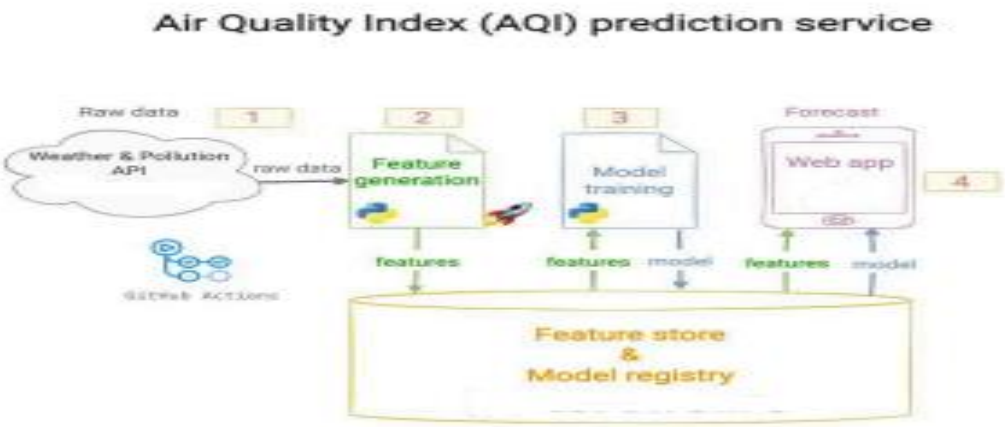
This page provides:

- This page explains Day-1 numeric AQI predictions using SHAP:
- Global SHAP: Overall feature importance
- Local SHAP: Feature contributions for the latest prediction

Explainability is limited to Day-1 regression models, as SHAP is most effective for continuous outputs.

Summary

The dashboard provides a complete, production-ready interface for AQI monitoring, forecasting, model governance, and explainability, aligned with real-time data pipelines and automated ML workflows.



5. Tools & Technologies Used

The Pearls AQI Predictor project leverages a mix of programming, data processing, machine learning, automation, and web technologies to build an end-to-end AQI forecasting system.

Category	Tools / Libraries
Programming Language	*Python*
Data Processing	*Pandas, NumPy*
Machine Learning	*Scikit-learn*
Visualization	*Matplotlib, Seaborn*
Model Explainability	*SHAP*
Web Application	*Streamlit,*
Automation / CI/CD	*GitHub Actions*
Data Sources / APIs	*Open Meteo , OpenWeather*
Version Control	*Git,Git Hub*
Feature Store	*Mongo DB Atlas*

6. Problems Faced During the Project

1. API Key Expiration

During development, the data source API key expired unexpectedly. This caused interruptions in real-time data ingestion and made continuous data collection difficult until the key was renewed.

2. Lack of Feature Store Knowledge (Initial Phase)

At the start of the project, the concept of a feature store was not fully understood. As a result, the entire pipeline was initially built using CSV files, and features were later loaded into the system from these files.

3. Incorrect Data Pipeline Approach

Using CSV files as the primary data source was a flawed approach for a real-time AQI prediction system. When the proper feature store–based architecture was adopted, the existing pipeline became incompatible.

4. Rebuilding the Project

Due to the incorrect initial design, the complete project had to be restructured and rebuilt using a live feature store and API-based ingestion, which increased development time and effort.

7. Conclusion

The Pearls AQI Predictor project delivers an end-to-end, automated system for forecasting air quality using real-time and historical data from OpenWeather and Open-Meteo. It combines feature engineering, machine learning, CI/CD automation, and SHAP-based explainability to provide reliable AQI forecasts.

Short-term AQI is predicted using regression, while longer-term forecasts rely on AQI class classification for stability. Overall, the system demonstrates practical use of data science and MLOps concepts in a scalable, production-ready air quality monitoring solution